SMART INDIA HACKATHON 2025

- **Problem Statement ID –25102**
- **Problem Statement Title- AI-based drop-out prediction and counseling system**
- **Theme-SMART AUTOMATION**
- **PS Category- Software**
- **Team ID- 66714**
- **Team Name: DataArt**

DataArt

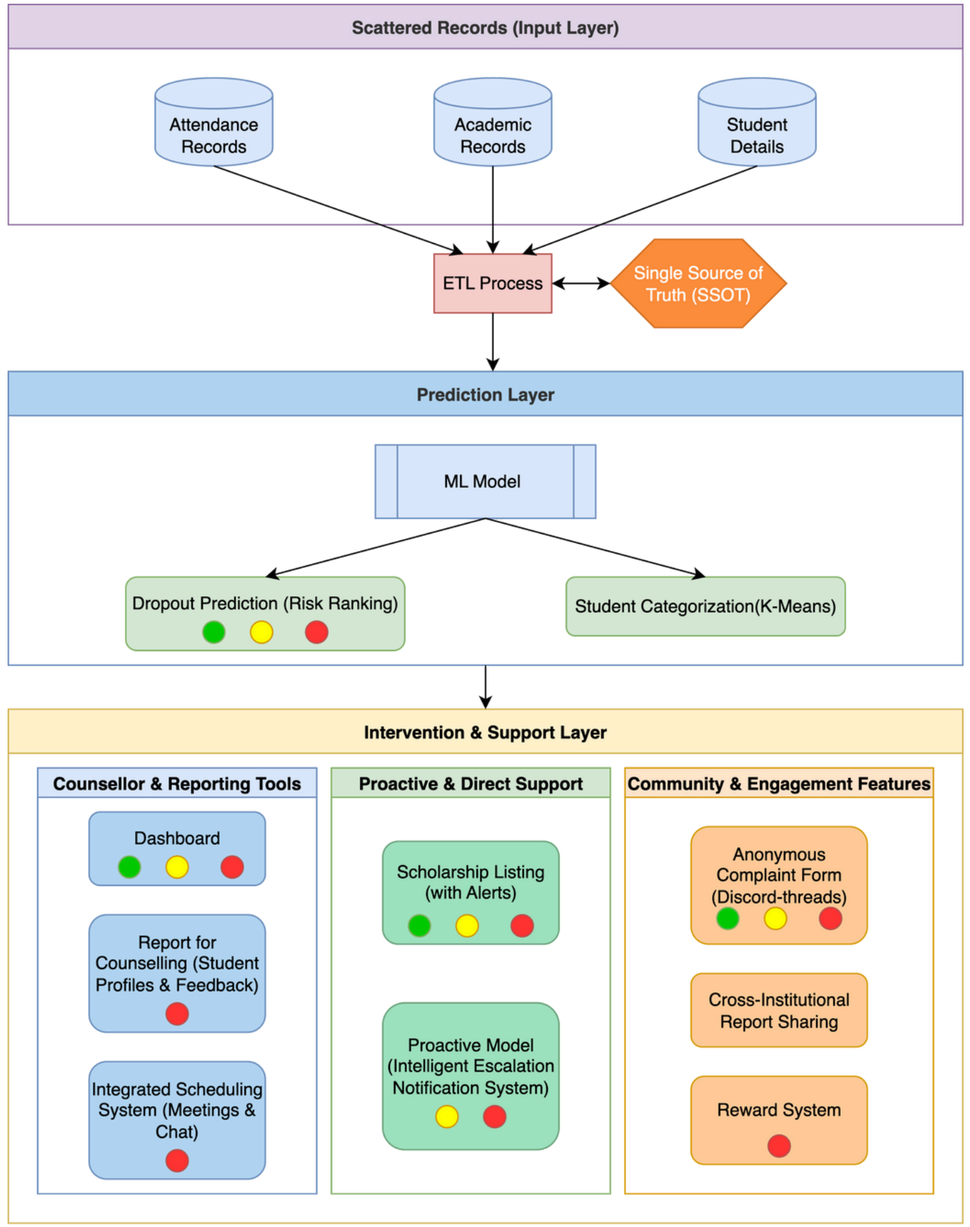
ACADELYTICS

Idea/Solution

- Our foundational layer uses **ETL** process to transform scattered records, like attendance and scores, into a **Single Source of Truth (SSOT)**.This unified data is structured around a **standardised schema**, allowing us to present a reliable view of each student's academic journey.
- **Dropout Prediction**: Utilises a **LightGBM** model on student data to accurately forecast dropout likelihood.
- **Risk Ranking**: Converts model predictions into a risk score, generating a prioritised "Top-K" list for counsellors.
- **Student Categorization:** Applies **K-Means clustering** to group students into actionable profiles (e.g., academic, financial ,psychology) for targeted support.
- **Dashboard:** flags prediction of students as a **color-coded system (red,yellow,green)**, charts and showing student-wise profile page.
- **Report for counselling**: student profile and **feedbacks** from parents and teachers collected through **personalised forms** on the student's situation.
- **Integrated Scheduling System:** Schedule meetings for student, mentor, and counsellor support.

Key Intervention & Unique Features:

- **Scholarship listing** for financially weak students based on their **eligibility criteria** with automatic **deadline alert.**
- **Proactive Model:** provides students showing early risk with resources to self-correct, and if factors like poor attendance or low subject grades escalate, an **Intelligent Escalation Notification System** sends multi-layered alerts to the entire network, simultaneously notifies administrators to review for potential teaching quality issues.
- An **anonymous complaint form** for students to safely voice **sensitive concerns** like bullying or mental distress with a feature like **Discord-threads** (e.g., #bullying, #academics) to identify & categorise feedbacks in risk indications.
- Sharing insights from **inter-college reports** on **observed patterns** and their **solutions**,enabling successful colleges to help struggling ones.
- Implementing **Reward System.**



DataArt

TECHNICAL APPROACH

SMART INDIA HACKATHON 2025

Frontend Technology

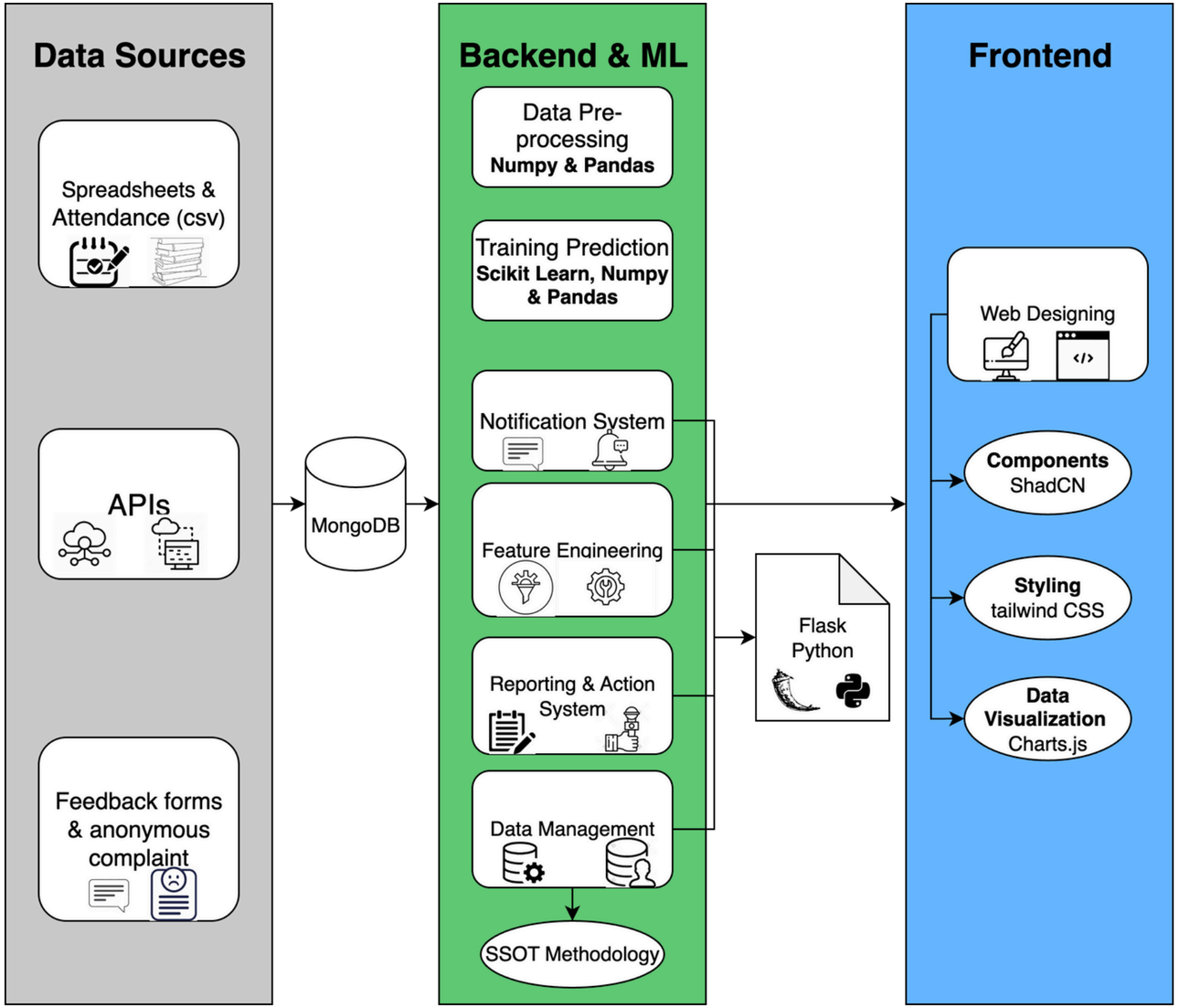
- **React**-Builds the user interface.
- **ShadCN**-Provides pre-built UI components.
- **Tailwind**-Styles the application's components.
- **Javascript,HTML,CSS**-Fundamental web building blocks.
- **Charts.js**-Creates in-app interactive charts.
- **Power BI**-External, high-level data reporting.

Backend Technology

- **Flask**-Builds the backend API.
- **Python**-Drives all backend logic.

AI & ML Technology

- **scikit learn**-Builds predictive machine learning models.
- **NumPy**-Performs fast numerical calculations.
- **Pandas**-Manages and analyzes data.
- **OtherTools**-ChatGPT 4.0,Gemini 2.5 Pro.



DataArt

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2025

Feasibility of ML Prediction

- Leverages proven machine learning techniques for **classification**, probability calibration to generate reliable risk scores, and clustering for meaningful **student categorization**.
- **Risk-based** interventions by identifying at-risk students early and enabling the **prioritization** of timely interventions.
- Deployment with lightweight, scalable and efficient models suitable for real-world deployment without heavy computational requirements.

Potential Challenges & Risks

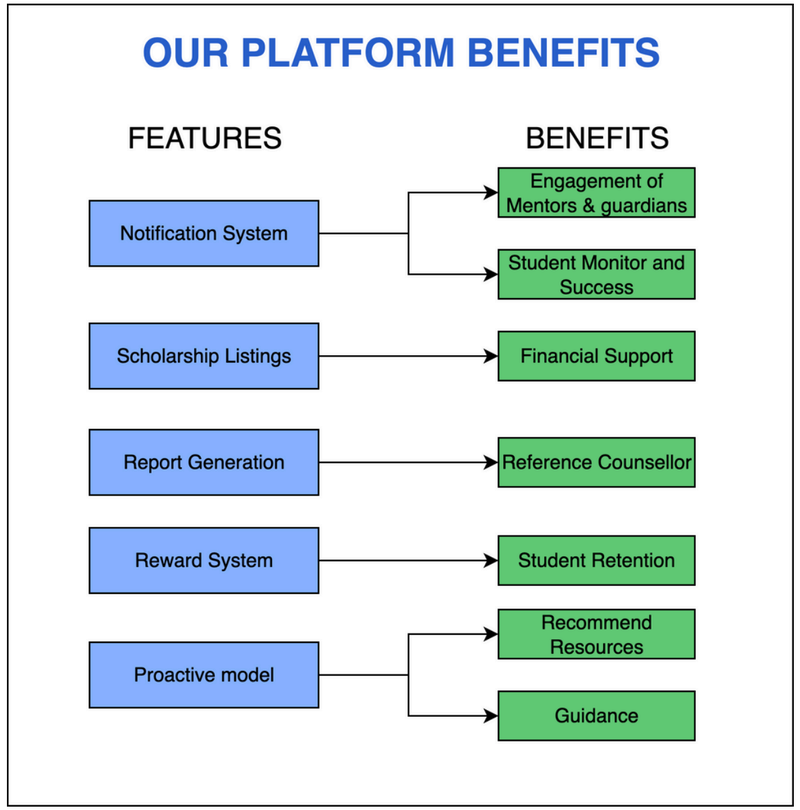
- **Class Imbalance**: Dropout cases are often much lower compared to non-dropouts, which may bias models toward majority outcomes.
- **Data Quality Issues**: Missing values, inconsistent entries, or noisy features could affect model performance.
- **Fairness Concerns**: Models may disproportionately flag certain demographic subgroups, raising risks of inequitable or biased outcomes.
- **Operational Adoption Risks**: Counselors and stakeholders may be hesitant to trust or integrate model outputs into existing decision-making workflows.

Strategies for overcoming the challenges	
Handle class imbalance	Apply class-weighted learning and evaluate metrics beyond accuracy to ensure minority/dropout cases are captured appropriately.
Prevent Label Leakage	Carefully select features that are available at the time of prediction snapshot, and avoid the use of future information not known at prediction time.
Feature Engineering	Craft trends, approximated ratios, respect_fail_count, and financial flags to boost model robustness.
Ensure Actionability	Present results as ranked-top K lists and simple prediction scores that are transparent, and practical for counselors to interpret and act upon.

DataArt

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2025



Potential Impact on Target Audience:

- **Decrease in Drop-out Rate-**The Platform will help in decreasing dropout rates both on state and national level.
- **Optimized Allocation of Public Funds-**By reducing costly student failure & repetition, taxpayers funds would help in building human capital rather than just managing failure.
- **Improvement in Educational Infrastructure-**The platform provides the core digital infrastructure to shift institutions from passive record-keeping to proactive, data-driven student support.

Benefits of the Solution:

1. Economic

- Improving the **Gross Enrolment Ratio** (GER).
- Maximizes the return on taxpayer investment in education.

2. Social

- Improves student well-being through early intervention.
- Increases parental and guardian engagement with automated alerts.

3. Technological

- Upgrades campus infrastructure from siloed data to an integrated, intelligent platform.
- Enables data-driven decision-making for administrators and policymakers by providing **intuitive dashboards** with **high-level statistics**.
- Creates a foundation for future AI applications in education.

4. Educational

- Shifts the educational model from reactive (post-failure) to **proactive** (early warning).
- Empowers educators with actionable insights, not just raw data.
- Allows for timely, personalized support tailored to specific student needs.
- Helps **identify and rectify** systemic issues in curriculum and teaching methods.

DataArt

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2025

Problems:
1)High annual dropout rates in India
2)Early Warning System by UNICEF

Existing Solutions:
go-redrock.com

Feasibility Study:
1)MANISH SISODIA SHIKSHA
2)Student Dropout Prediction-pmc.ncbi.nlm.nih.gov

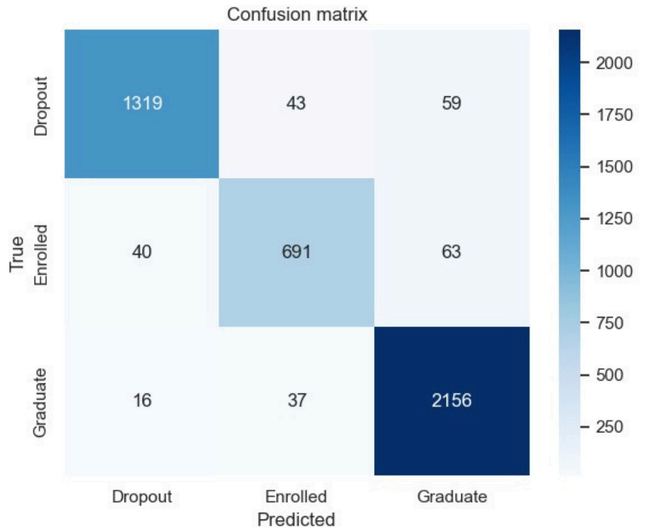
Our Key Findings

Dataset Used:
1)UCI Machine Learning Repository – Higher Education Students Performance Dataset (Portuguese)
Dataset of **4,424 students** with demographic, academic, and socio-economic features, labeled with outcomes: Dropout, Enrolled, or Graduate.

Reference Paper:
1)**Early Prediction of Student's Performance in Higher Education:** A Case Study
Authors: Mónica V. Martins, Daniel Tolledo, Jorge Machado, Luís M. T. Baptista, and Valentim Realinho
Year: 2021 — Published in Trends and Applications in Information Systems and Technologies

More References:
1)Moodle - **Learning Management System**
2) EAB - **Early Alert, Scheduling, and Case Management.**

Comparison With Existing Solutions

Feature	Our System	Existing System
Reward System	✓	✗
Intelligent Escalation System	✓	✗
Proactive Model	✓	✓
Personalised Scholarship Listing	✓	✗
Cross-Institute Report Sharing	✓	✗
Anonymous Complaint Forms	✓	✗
Scheduling System	✓	✓

Performance of Our Model